

5A) 1. Calculate the fugacity of water at { T = 250 C and P = 70 bar } with Steam Tables.

$$dG = RT d\ln f \rightarrow \int_{G_1}^{G_2} dG = \int_{f_1}^{f_2} RT d\ln f \quad T \text{ is constant} \rightarrow G_2 - G_1 = RT \ln \frac{f_2}{f_1} \quad (1)$$

We know that at a very low temperature, the fugacity is equal to the pressure: $f_1 = P_1$ since at a very low pressure we can assume an ideal gas behavior.

We choose a very small pressure $P_1 = 0.01 \text{ bar} = f_1$

The state at which $T_1 = 250^\circ\text{C}$, $P_1 = 0.01 \text{ bar}$ serves as a reference point to calculate the fugacity at the desired state of $T_2 = 250^\circ\text{C}$, $P_2 = 70 \text{ bar}$

We use the calculator's function, HSTEAM, to get the molar enthalpy at state 1 and 2:

$$H_1 = \text{HSTEAM}(250, 0.01) = 53607.9 \frac{\text{kJ}}{\text{kmol}}$$

$$H_2 = \text{HSTEAM}(250, 70) = 19546.8 \frac{\text{kJ}}{\text{kmol}}$$

Note that since the saturation pressure at 250°C is 39.78 bar, State 1 must be in the vapor phase and State 2 must be in the liquid phase.

We use the calculator's function SSTEAM to get the molar entropy at State 1 and 2:

$$S_1 = \text{SSTEAM}(250, 0.01) = 182.98 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

$$S_2 = \text{SSTEAM}(250, 70) = 50.16 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

Now that we have enthalpy and entropy, we can calculate Gibbs free energy using the relation of State 1 and 2:

$$G_1 = H_1 - TS_1 = 53607.9 \frac{\text{kJ}}{\text{kmol}} - (250 + 273.15) \text{K} \cdot \left(182.98 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \right) = -42119.7 \frac{\text{kJ}}{\text{kmol}}$$

$$G_2 = H_2 - TS_2 = 19546.8 \frac{\text{kJ}}{\text{kmol}} - (250 + 273.15) \text{K} \cdot \left(50.16 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \right) = -6694.49 \frac{\text{kJ}}{\text{kmol}}$$

We rearrange equation 1 to solve for f_2 :

$$f_2 = f_1 \cdot \exp\left(\frac{G_2 - G_1}{RT}\right) = (0.01 \text{ bar}) \cdot \exp\left(\frac{-6694.5 \frac{\text{kJ}}{\text{kmol}} + 42119.7 \frac{\text{kJ}}{\text{kmol}}}{8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \cdot (250 + 273.15) \text{K}}\right) = 34.45 \text{ bar}$$

→ The fugacity of water at $T=250^\circ\text{C}$ and $P=70 \text{ bar}$ is $f = 34.45 \text{ bar}$

D8	f_x	=EXP((D7-C7)/((8.314)*(D3+273.15)))*0.01
----	-------	--

	A	B	C	D
1	5A 1)			
2	Conditions and Properties	Units	State 1	State 2
3	Temperature	C	250.0000	250.0000
4	Pressure	bar	0.0100	70.0000
5	Molar Enthalpy	kJ/kmol	53607.9038	19546.7779
6	Molar Entropy	kJ/(kmol K)	182.9832	50.1601
7	Gibbs Free Energy	kJ/kmol	-42119.7475	-6694.4851
8	Fugacity (f)	bar	0.0100	34.4518

2. Assuming the water follows the Peng-Robinson Equation of State both in the gas and liquid phases.

Peng-Robinson EOS: $P + \frac{a\sigma}{V(V+b)+b(V-b)} = \frac{RT}{V-b}$ where $a=0.45723553 \frac{(RT_c)^2}{P_c}$,

$$b=0.07779607 \frac{RT_c}{P_c}, \sigma = [1 + \kappa(1 - \sqrt{T_r})]^2, A = \frac{a\sigma P}{(RT)^2} \text{ and } B = \frac{bP}{RT}$$

We start by finding the Z-form of the Peng-Robinson EOS:

We multiply both sides by $\frac{V}{RT} \rightarrow \frac{PV}{RT} + \frac{a\sigma V}{RT[V(V+b)+b(V-b)]} = \frac{V}{V-b}$

$$Z = \frac{PV}{RT}, \text{ we define } A = \frac{a\sigma P}{(RT)^2} \text{ and } B = \frac{bP}{RT} \rightarrow BV = \frac{bPV}{RT} \rightarrow \frac{B}{Z} = \frac{b}{V}$$

$$\rightarrow Z + \frac{a\sigma V}{V^2 RT \left[\left(1 + \frac{b}{V}\right) + \frac{b}{V} \left(1 - \frac{b}{V}\right) \right]} = \frac{1}{1 - \frac{b}{V}} \rightarrow Z + \frac{a\sigma P}{V(RT)^2 \left[\left(1 + \frac{b}{V}\right) + \frac{b}{V} \left(1 - \frac{b}{V}\right) \right]} \cdot \frac{RT}{P} = \frac{1}{1 - \frac{b}{V}}$$

We rewrite it in terms of A & B:

$$Z + \frac{A}{\left(1 + \frac{B}{Z}\right) + \frac{B}{Z} \left(1 - \frac{B}{Z}\right)} \cdot \frac{RT}{PV} = \frac{1}{1 - \frac{B}{Z}} \rightarrow Z + \frac{A}{Z \left[\left(1 + \frac{B}{Z}\right) + \frac{B}{Z} \left(1 - \frac{B}{Z}\right) \right]} = \frac{1}{1 - \frac{B}{Z}}$$

$$Z + \frac{A}{Z \left[\frac{1}{Z}(Z+B) + \frac{B}{Z^2}(Z-B) \right]} = \frac{Z}{Z-B} \rightarrow Z + \frac{A}{(Z+B) + \frac{B}{Z}(Z-B)} = \frac{Z}{Z-B}$$

$$\rightarrow Z(Z-B) \left[Z + 2B - \frac{B^2}{Z} \right] + A(Z-B) - Z \left[Z + 2B - \frac{B^2}{Z} \right] = 0$$

$$\rightarrow Z^3 + 2BZ^2 - B^2Z - BZ^2 - 2B^2Z + B^3 + AZ - AB - Z^2 - 2BZ + B^2 = 0$$

$$\rightarrow Z^3 + BZ^2 - 3B^2Z + B^3 + AZ - AB - Z^2 - 2BZ + B^2 = 0$$

$$\rightarrow \text{The Z-form: } Z^3 + Z^2(B-1) + Z(A-2B-3B^2) - AB + B^2 + B^3 = 0$$

Solving this cubic equation, we should get two roots at T=250°C and P=70 bar.

We can also use the ZCOMP function to get these roots:

$$Z_v = \text{ZCOMP}(\text{H}_2\text{O}, 250, 70, 1) = 0.69882$$

$$Z_v = \text{ZCOMP}(\text{H}_2\text{O}, 250, 70, 1) = 0.04488$$

We find the Z,ρ-Form of the Peng-Robinson EOS:

$$\frac{PV}{RT} + \frac{a\sigma V}{RT[V(V+b)+b(V-b)]} = \frac{V}{V-b} \rightarrow Z = \frac{1}{1-b\rho} - \frac{a\sigma}{RTV[1+2b\rho-b^2\rho^2]}$$

$$\rightarrow Z = \frac{1}{1-b\rho} - \frac{a\sigma\rho}{RT[1+2b\rho-b^2\rho^2]}$$

$$\ln \phi = Z - 1 - \ln Z + \int_0^\rho (Z-1) \frac{d\rho}{\rho} ; \text{ we plug the Z,}\rho\text{-Form inside the integral:}$$

$$\ln \phi = Z - 1 - \ln Z + \int_0^{\rho} \left(\frac{1}{\rho - b\rho^2} - \frac{a\sigma}{RT[1 + 2b\rho - b^2\rho^2]} - \frac{1}{\rho} \right) d\rho$$

We integrate this function and we take the Exp of both side of the equation to get the fugacity coefficient of the PR EOS.

This can also be achievable using the function PHIC in the calculator to get two fugacity coefficients for both Z_L and Z_V :

$$\Phi_V = \text{PHIC}(\text{H}_2\text{O}, \text{H}_2\text{O}, 250, 70, 0.69882) = 0.77156$$

$$\Phi_L = \text{PHIC}(\text{H}_2\text{O}, \text{H}_2\text{O}, 250, 70, 0.04488) = 0.50763$$

Since $\phi = \frac{f}{P}$ then $f = \phi P$:

$$\rightarrow f_V = \phi_V P = 0.77156 \times 70 \text{ bar} = 54.0094 \text{ bar}$$

$$f_L = \phi_L P = 0.50763 \times 70 \text{ bar} = 35.534 \text{ bar}$$

Since we have two fugacities at $T=250^\circ\text{C}$ and $P = 70 \text{ bar}$, then the lower fugacity represents the more stable phase.

$\rightarrow$ the fugacity of water is $f_L = 35.534 \text{ bar}$ and it is in the liquid phase.

Vapor Phase		
Molar Composition		
New Molar Composition		
Compressibility Factor		0.69882
Fugacity Coefficient		0.77156
Saturated Fugacity Coefficient		
Fugacity, bar		54.00944
Molar Enthalpy, kJ/kmol		19546.77790
Molar Entropy, kJ/(kmol K)		
Molar Density, kmol/m ³		
Molar Heat Capacity, kJ/(kmol K)		
Liquid Phase		
Molar Composition		
New Molar Composition		
Compressibility Factor		0.04488
Fugacity Coefficient		0.50763
Liquid Activity Coefficient		
Fugacity, bar		35.53382
Molar Enthalpy, kJ/kmol		19546.77790
Molar Entropy, kJ/(kmol K)		50.16011

3. Assuming the water follows the Peng-Robinson Equation of State only in the gas phase.

Since water only follows the PR EOS in the gas phase only and water is liquid at $T = 250^\circ\text{C}$ and $P = 70$ bar, we need to find an equation to calculate the liquid fugacity, the Poynting Method:

$$dG = -SdT + VdP \quad \text{under constant Temperature} \rightarrow dG = VdP \quad (1)$$

We also know that $G = \text{Constant} + RT \ln f$ by taking the derivative of both sides:

$$dG = RT d \ln f \quad (2)$$

by equating (1) and (2): $RT d \ln f = VdP$ we integrate both sides:

$$\int_{f_L^{SAT}}^{f_L} d \ln f = \int_{P^{SAT}}^P \frac{V}{RT} dP \quad \text{note that } V \text{ is constant because in incompressible liquids } V_L = V_L^{SAT}$$

$$\rightarrow \ln \frac{f_L}{f_L^{SAT}} = \frac{V_L}{RT} (P - P^{SAT}) \rightarrow f_L = f_L^{SAT} \exp \left(\frac{V_L}{RT} (P - P^{SAT}) \right) \quad (\text{Poynting Method})$$

At saturation point, $f_L^{SAT} = f_V^{SAT}$ and since we are assuming that water follows PR EOS in the gas phase, we can use the PHIC function in the calculator to find f_V^{SAT} :

First we find the saturation pressure at 250°C using the EOSPsat function:

$$\text{EOSPsat}(\text{H}_2\text{O}, 250, 70) = 40.119 \text{ bar.}$$

Note that we put 70 bar as a guess value for the EOSPsat function.

We calculate the compressibility factor Z_V^{SAT} at 250°C and 40.1 bar by solving the cubic form of the PR EOS; we use the ZCOMP function in the calculator:

$$\text{ZCOMP}(\text{H}_2\text{O}, 250, 40.119, 1) = 0.8507$$

Using the PHIC function in the calculator we can find ϕ_V^{SAT} at 250°C and 40.1 bar:

$$\phi_V^{SAT} = \text{PHIC}(\text{H}_2\text{O}, \text{H}_2\text{O}, 250, 40.119, 0.8507) = 0.8689$$

$$\rightarrow f_L^{SAT} = f_V^{SAT} = \phi_V^{SAT} P^{sat} = 0.8689 \times 40.119 \text{ bar} = 34.858 \text{ bar}$$

To calculate V_L we use the Rackett equation:

$$V_L = V_L^{SAT} = \frac{RT_C}{P_C} Z_C^{1+(1-T_r)^{2/7}}$$

We use the Rackett function in the calculator to get this value:

$$\rho_L = \rho_L^{SAT} = \text{Rackett}(\text{H}_2\text{O}, 250) = 44.87 \frac{\text{kmol}}{\text{m}^3} \rightarrow V_L = \frac{1}{\rho_L} = 0.0223 \frac{\text{m}^3}{\text{kmol}}$$

We plug all these values in the Poynting Equation to get the fugacity of liquid water at 250°C and 70 bar:

$$f_L = f_L^{SAT} \exp \left(\frac{V_L}{RT} (P - P^{SAT}) \right) = 34.858 \text{ bar} \times \exp \left(\frac{0.0223 \frac{\text{m}^3}{\text{kmol}}}{8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}} \left(70 \text{ bar} \frac{100 \text{ kPa}}{\text{bar}} - 40.119 \text{ bar} \frac{100 \text{ kPa}}{\text{bar}} \right) \right)$$

$$\rightarrow \boxed{f_L = 35.3958 \text{ bar}}$$

C11		f_x	$=C8*EXP((C10*(C4*100-C5*100))/(8.314*(C3+273.15)))$			
	A	B	C	D	E	F
1	5A 3)					
2	Conditions and Properties	Units				
3	Temperature	C	250.0000			
4	Pressure	bar	70.0000			
5	Saturation Pressure (PR-EOS)	bar	40.11923249	EOSPSAT		
6	Compressibility Factor SAT Vapor		0.8507			
7	Fugacity Coefficient SAT Vapor		0.8689			
8	Saturated Vapor fugacity	bar	34.8580			
9	Molar Density	kmol/m3	44.8726			
10	Molar Volume	m3/kmol	0.0223			
11	Fugacity (f) Liquid	bar	35.3958			

Percent Error;

Since the steam table method gives the experimental data value of the fugacity, we calculate the percent error of the other two methods against the steam table.

$$\text{Method 2: } \% \text{ Error} = \left| \frac{35.53382 - 34.4518}{34.4518} \right| \times 100\% = 3.14\%$$

$$\text{Method 3: } \% \text{ Error} = \left| \frac{35.3958 - 34.4518}{34.4518} \right| \times 100\% = 2.74\%$$

We notice that the 3rd method using the Poynting equation gives a more accurate result for the fugacity of liquid water than assuming PR EOS.

5B) 1. Verify whether propane is in the liquid or gas phase or whether there are two phases present. Assume propane follows the van der Waals equation of state.

To figure out if propane is liquid or vapor at $T = 50^\circ\text{C}$ and $P = 5$ bar assuming it follows Van der Waals EOS, we find the liquid and vapor fugacity coefficients and whichever is lower corresponds to the stable phase:

First we calculate the vapor and liquid compressibility factor by solving the cubic form of the Van der Waals EOS. To get these values we use the ZCOMP function in the calculator:

$$Z_V = ZCOMP(C_3H_8, 50, 5, 1) = 0.94954$$

$$Z_L = ZCOMP(C_3H_8, 50, 5, 1) = 0.94954$$

Then we use the PHIC function in the calculator to get the liquid and vapor fugacity coefficients:

$$\phi_V = PHIC(C_3H_8, C_3H_8, 50, 5, 0.94954) = 0.95189$$

$$\phi_L = PHIC(C_3H_8, C_3H_8, 50, 5, 0.94954) = 0.95189$$

Since both the fugacity coefficients are the same because $Z_V = Z_L$, it means there is one real root for the cubic equation and propane is either liquid or vapor at $T = 50^\circ\text{C}$ and $P = 5$ bar and it cannot be in a saturated phase.

To figure out which phase it is, we find the saturation pressure of propane at $T = 50^\circ\text{C}$ using the calculator's function EOSPsat:

First we need an initial guess for the saturation pressure, we use the Psat function calculates ideal saturation pressure according to the Antoine's Equation:

$$P_0 = Psat(C_3H_8, 50) = 15.76 \text{ bar}$$

$$\rightarrow P_V^{Sat} = EOSPsat(C_3H_8, 50, 15.76) = 24.188 \text{ bar}$$

Since $P = 5 \text{ bar} < 24.188 \text{ bar}$ then propane is in the gas phase at $T = 50^\circ\text{C}$ and $P = 5 \text{ bar}$

Temperature, C	50.00000
Pressure, bar	5.00000
Vapor Phase	
Compressibility Factor	0.94954
Fugacity Coefficient	0.95306
Liquid Phase	
Compressibility Factor	0.94954
Fugacity Coefficient	0.95189
Psat	15.75972
EOSPsat	24.18834

2. Calculate the numerical value of the derivative $\left(\frac{\partial T}{\partial P}\right)_S$ assuming propane is ideal gas.

We start with $H=U+PV$ we take the differential of it: $dH=dU+PdV+VdP$ (1)
however, $dU=\delta Q+\delta W$ (2)

In a reversible system, there is no increase in entropy $\rightarrow dS=\frac{\delta Q}{T} \rightarrow \delta Q=TdS$

We also know: $\delta W=-PdV$

We substitute in (2) $\rightarrow dU=TdS-PdV$ and we substitute this in (1):

$$dH=TdS-PdV+PdV+VdP=TdS+VdP \quad (3)$$

We need to find a way to write $\left(\frac{\partial T}{\partial S}\right)_P$ in an easier to calculate way:

$$H(T,P) \rightarrow dH=\left(\frac{\partial H}{\partial T}\right)_P dT+\left(\frac{\partial H}{\partial P}\right)_T dP ; \text{ we know that } \left(\frac{\partial H}{\partial T}\right)_P=C_p$$

We divide both sides of (3) by dP while keeping T constant:

$$\rightarrow \left(\frac{\partial H}{\partial P}\right)_T=T\left(\frac{\partial S}{\partial P}\right)_T+V$$

$$\rightarrow dH=C_p dT+\left(T\left(\frac{\partial S}{\partial P}\right)_T+V\right)dP \quad (4)$$

Next we want to write $\left(\frac{\partial S}{\partial P}\right)_T$ in a different form to make it easier to calculate:

We know that Gibbs Free Energy: $G=H-TS$ we take the derivative of it:

$$\rightarrow dG=dH-TdS-SdT \rightarrow dG=TdS+VdP-TdS-SdT=VdP-SdT \text{ by taking the second}$$

$$\text{derivative we find that: } \left(\frac{\partial S}{\partial P}\right)_T=-\left(\frac{\partial V}{\partial T}\right)_P$$

$$\text{We substitute in (4): } dH=C_p dT+\left(-T\left(\frac{\partial V}{\partial T}\right)_P+V\right)dP \quad (5)$$

$$\text{We call } \beta=\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P \text{ we substitute in (5): } dH=C_p dT+(1-\beta T)VdP \quad (6)$$

$$\text{We substitute (6) in (3): } TdS+VdP=C_p dT+(1-\beta T)VdP$$

$$\rightarrow dS=C_p \frac{dT}{T}-\beta VdP \quad (7)$$

To calculate the value of $\left(\frac{\partial T}{\partial P}\right)_S$ we divide both sides of Eq (7) by dP while keeping

Entropy constant:

$$0=\frac{C_p}{T}\left(\frac{\partial T}{\partial P}\right)_S-\beta V \rightarrow \left(\frac{\partial T}{\partial P}\right)_S=\frac{\beta VT}{C_p} \quad (8)$$

We calculate β of an ideal gas:

$$V=\frac{RT}{P}=\frac{8.3145 \frac{kJ}{kmol K}(50+273.15)K}{500 kPa}=5.374 \frac{m^3}{kmol}$$

$$\rightarrow \left(\frac{\partial V}{\partial T} \right)_P = \frac{R}{P} \rightarrow \beta = \frac{R}{PV} = \frac{8.3145 \frac{\text{kJ}}{\text{kmol K}}}{500 \text{ kPa} \times 5.374 \text{ m}^3} = 0.003095 \frac{1}{\text{K}}$$

We use the CPIG function in the calculator to get the heat capacity value:

$$C_p = \text{CPIG}(C_3H_8, 50) = 79.26 \frac{\text{kJ}}{\text{kmol K}}$$

$$\rightarrow \left(\frac{\partial T}{\partial P} \right)_S = \frac{\beta V T}{C_p} = \frac{0.003095 \frac{1}{\text{K}} \times 5.374 \frac{\text{m}^3}{\text{kmol}} \times (50 + 273.15) \text{ K}}{79.26 \frac{\text{kJ}}{\text{kmol K}}} \times 100 \frac{\text{kPa}}{\text{bar}} = 6.779487 \frac{\text{K}}{\text{bar}}$$

→ the numerical value of the derivative assuming propane is ideal gas:

$$\left(\frac{\partial T}{\partial P} \right)_S = 6.779487 \frac{\text{K}}{\text{bar}}$$

C9					=100*(C5*C8*(C3+273.15))/C6	
	A	B	C	D	E	
1	5B 2					
2	Properties	Units	Values			
3	Temperature	K	50			
4	Pressure	bar	5			
5	Beta	1/K	0.003095			
6	Cp	kJ/kmol K	79.26355			
7	R	kJ/kmol K	8.3145			
8	V	m^3/kmol	5.373661			
9	dT/dP	K/bar	6.779487			

3. Calculate the numerical value of the derivative $\left(\frac{\partial T}{\partial P}\right)_s$ by using first order numerical differencing and assuming that propane follows the van der Waals equation of state.

We perform numerical differencing using the hint formula in the assignment

$$\frac{dT}{dP} \approx \frac{T(P + \Delta P) - T(P)}{\Delta P}$$

We start at $T_1 = 50^\circ\text{C}$ and $P_1 = 5$ bar and we use the ZCOMP function to get the vapor compressibility factor at this state and we make sure to select the Van der Waals option in the calculator:

$$Z_{v1} = \text{ZCOMP}(C_3H_8, 50, 5, 1) = 0.94954$$

We evaluate the entropy using the entropy function:

$$S_1 = \text{ENTROPY}(C_3H_8, 50, 5, 0.94954) = -277.447 \frac{\text{kJ}}{\text{kmol K}}$$

Next we pick a very small change in $\Delta P = 0.0001 \rightarrow P_2 = 5.0001$ bar

We pick a guess value for $T_2 = 52^\circ\text{C}$

We evaluate Z_{v2} and S_2 at the second state:

$$Z_{v2} = \text{ZCOMP}(C_3H_8, 50, 5.0001, 1) = 0.95031$$

$$S_2 = \text{ENTROPY}(C_3H_8, 50, 5.0001, 0.95031) = -276.949 \frac{\text{kJ}}{\text{kmol K}}$$

Since derivative is under constant Entropy, we use solver to set $S_1 = S_2$ by changing T_2 and we get $T_2 = 50.00072^\circ\text{C}$

Now that we have T_2 at $P_2 = 5.0001$ bar we use the differencing formula to get the value of the derivative:

$$\left(\frac{\partial T}{\partial P}\right)_s = \frac{50.00072^\circ\text{C} - 50^\circ\text{C}}{5.0001 \text{ bar} - 5 \text{ bar}} = 7.171964 \frac{\text{K}}{\text{bar}}$$

→ **the numerical value of the derivative by using first order numerical differencing is**

$$\boxed{\left(\frac{\partial T}{\partial P}\right)_s = 7.171964 \frac{\text{K}}{\text{bar}}}$$

C9		f_x	=(C4-D4)/(C5-D5)		
1	2	A	B	C	D
1	VLE Calculations				
2	Fluid Conditions		Mixture	C3H8	
3	Molar Composition				
4	Temperature, C			50.00000	50.00072
5	Pressure, bar			5.00000	5.00010
6					
7	Saturated Vapor Pressure, bar			24.18834	
8	Saturated Vapor Temperature, C				
9	dT/dP		!	7.171964	
10	K-Values				
11	New K-Values				
12					
13	Vapor to Feed Molar Ratio [a=V/F]				
14	The Rachford-Rice Equation				
15	Discriminant of the Cubic EOS				
16	Vapor Phase				
17	Molar Composition				
18	New Molar Composition				
19	Compressibility Factor			0.94934	0.94934
20	Fugacity Coefficient				
21	Saturated Fugacity Coefficient				
22					
23	Molar Enthalpy, kJ/kmol				
24	Molar Entropy, kJ/(kmol K)			-277.44705	-277.44705

4. Develop an expression for the derivative $\left(\frac{\partial T}{\partial P}\right)_s$ using the van der Waals equation of state and calculate its numerical value.

We use the same formula we derived in part 2 $\left(\frac{\partial T}{\partial P}\right)_s = \frac{\beta VT}{C_p}$ but this time under Van der Waals EOS.

We use ZCOMP function to get the value of Z:

$$Z_V = ZCOMP(C_3H_8, 50, 5, 1) = 0.9495$$

To calculate the value of β we use the ZPROP function with the index 5:

$$\beta = ZPROP(5, C_3H_8, 50, 5, 0.9495) = 0.003503 \frac{1}{K}$$

To calculate the heat capacity, we use the function ZPROP with the index 2:

$$C_p = ZPROP(C_3H_8, 50, 5, 0.9495) = 80.5303 \frac{kJ}{kmol K}$$

To calculate the molar volume we use the Density function and we take the inverse of it:

$$V = \frac{1}{DENSITY(50, 5, 0.9495)} = 5.1025 \frac{m^3}{kmol}$$

We plug all these values into the derivative formula:

$$\left(\frac{\partial T}{\partial P}\right)_s = \frac{\beta VT}{C_p} = \frac{0.003503 \frac{1}{K} \times 5.1025 \frac{m^3}{kmol} \times (50 + 273.15) K}{80.5303 \frac{kJ}{kmol K}} = 7.172031$$

We find that the numerical value of $\left(\frac{\partial T}{\partial P}\right)_s = 7.172031 \frac{K}{bar}$

C10		fx		=100*(C6*C9*(C3+273.15))/C7		
	A	B	C	D	E	
1	5B 4					
2	Properties	Units	Values			
3	Temperature	K	50			
4	Pressure	bar	5			
5	Z		0.949539			
6	Beta	1/K	0.003503			
7	Cp	kJ/kmol K	80.53046			
8	R	kJ/kmol K	8.3145			
9	V	m^3/kmol	5.102503			
10	dT/dP	K/bar	7.172031			

5. Prepare a table that shows the results of the three methods.

Part Number	$\left(\frac{\partial T}{\partial P}\right)_s \left[\frac{K}{\text{bar}}\right]$
Part number 2	6.779487
Part number 3	7.171964
Part number 4	7.172031